

# **Intelligent Literature Review System (ILRS)**

## **Introduction**

The Intelligent Literature Review System (ILRS) is an AI-powered platform designed to help users efficiently search, analyze, and summarize research papers. The system leverages natural language processing and machine learning techniques to provide relevant academic results, generate summaries, and identify research gaps.

## **OBJECTIVE**

The main objective of this system is to simplify the literature review process by:

- Providing fast research paper search
- Generating AI-based summaries
- Identifying similarities and research gaps
- Allowing users to manage and organize research papers

## KEY FEATURES

- ★  Smart Paper Search (AI-based)
- ★  AI Summary Generation
- ★  Comparative Analysis
- ★  Research Gap Detection
- ★  Upload PDF / ZIP papers
- ★  User Authentication (Login/Signup)
- ★  Subscription Plans
- ★  Dashboard Analytics

## **HOW SYSTEM WORKS**

The system follows a simple workflow:

1. User logs into the system
2. User searches research papers
3. System retrieves relevant papers using embeddings
4. User can generate AI summaries
5. User can analyze multiple papers
6. System identifies research gaps

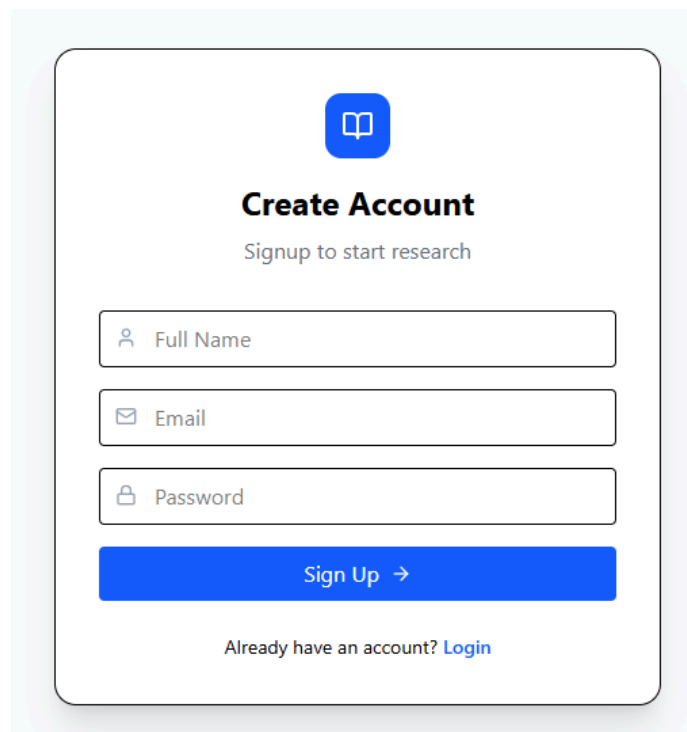
# Sign Up & Login

Users must create an account to access features.

## Steps:

1. Go to Sign Up page
2. Enter name, email, password
3. Click Sign Up
4. Login using credentials

## Signup page





### Create Account

Signup to start research

 Full Name

 Email

 Password

Sign Up →

Already have an account? [Login](#)

## Login page



### Welcome Back

Sign in to continue

 Email

 Password

Login →

Don't have an account? [Sign Up](#)

# SEARCH MODULE

Users can search research papers using keywords.

## Steps:

1. Enter search query
2. Click Search
3. View results
4. Open paper or summarize

Search Papers

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Plan

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Q machine learning in agriculture

Q Search

**Semantic Referee: A Neural-Symbolic Framework for Enhancing Geospatial Semantic Segmentation**  
Similarity Score: 0.403  
Understanding why machine learning algorithms may fail is usually the task of the human expert that uses domain knowledge and contextual information to discover systematic shortcomings in either the data or the algorithm. In this paper, we propose a semantic referee, which is able to extract qualitative features of the errors emerging from deep machine learning frameworks and suggest corrections. The semantic referee relies on ontological reasoning about spatial knowledge in order to characterize errors in terms of their spatial relations with the environment. Using semantics, the reasoner interacts with the learning algorithm as a supervisor. In this paper, the proposed method of the interaction between a neural network classifier and a semantic referee shows how to improve the performance of semantic segmentation for satellite imagery data.  
[AI Summary](#) [Open Paper](#) [Save to Library](#)

**Breiman's "Two Cultures" Revisited and Reconciled**  
Similarity Score: 0.353  
In a landmark paper published in 2001, Leo Breiman described the tense standoff between two cultures of data modeling: parametric statistical and algorithmic machine learning. The cultural division between these two statistical learning frameworks has been growing at a steady pace in recent years. What is the way forward? It has become blatantly obvious that this widening gap between "the two cultures" cannot be averted unless we find a way to blend them into a coherent whole. This article presents a solution by establishing a link between the two cultures. Through examples, we describe the challenges and potential gains of this new integrated statistical thinking.  
[AI Summary](#) [Open Paper](#) [Save to Library](#)



# AI SUMMARY

The system provides automatic summaries using AI.

## Steps:

1. Click "AI Summary"
2. Summary appears instantly

The screenshot shows a web application interface. At the top left is the text 'Search Papers'. At the top right is a user profile for 'tanishtha' with a 'Plan' button and a circular avatar containing the letter 't'. Below the header is a search bar with the text 'machine learning in agriculture' and a blue 'Search' button. The main content area displays a search result for a paper titled 'Semantic Referee: A Neural-Symbolic Framework for Enhancing Geospatial Semantic Segmentation' with a 'Similarity Score: 0.403'. The paper's abstract is visible. Below the abstract are three links: 'AI Summary' (highlighted with a red bracket), 'Open Paper', and 'Save to Library'. A light blue box contains the AI-generated summary of the paper, which is also highlighted with a red bracket.

Search Papers

tanishtha  
Plan

Q machine learning in agriculture Search

**Semantic Referee: A Neural-Symbolic Framework for Enhancing Geospatial Semantic Segmentation**  
Similarity Score: 0.403

Understanding why machine learning algorithms may fail is usually the task of the human expert that uses domain knowledge and contextual information to discover systematic shortcomings in either the data or the algorithm. In this paper, we propose a semantic referee, which is able to extract qualitative features of the errors emerging from deep machine learning frameworks and suggest corrections. The semantic referee relies on ontological reasoning about spatial knowledge in order to characterize errors in terms of their spatial relations with the environment. Using semantics, the reasoner interacts with the learning algorithm as a supervisor. In this paper, the proposed method of the interaction between a neural network classifier and a semantic referee shows how to improve the performance of semantic segmentation for satellite imagery data.

[AI Summary](#) [Open Paper](#) [Save to Library](#)

Here is a summary of the research paper: \* \*\*Proposed "Semantic Referee":\*\* Developed a framework that utilizes \*\*ontological reasoning\*\* and \*\*spatial knowledge\*\* to automatically identify, characterize, and suggest corrections for qualitative errors in deep learning models. \* \*\*Supervisory Methodology:\*\* The system acts as a supervisor to a neural network, analyzing \*\*spatial relations\*\* between detected objects and their environment to detect systematic shortcomings that the algorithm might miss. \* \*\*Improved Accuracy:\*\* The study demonstrates that integrating this semantic reasoning with a neural network classifier significantly enhances the performance of \*\*semantic segmentation in satellite imagery\*\*.

## RESULTS & ANALYSIS

Users can compare multiple papers to identify patterns and gaps.

### Steps:

1. Select 2+ papers
2. Click "Run Analysis"
3. View insights

 ILR System



 Dashboard

 Search Papers

 Upload Papers

 Results & Analysis

 AI Research Assistant

 Profile

 Subscription

 Help & Support

 Contact Us

 Log out

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Results & Analysis

Select multiple papers to perform deep AI analysis and gap detection.

 Run Comparative Analysis

 Clear

 Your Library (2)

**Semantic Referee: A Neural-Symbolic Framework for...**  
2024 • arXiv

**Overview of FPGA deep learning acceleration based on...**  
2024 • arXiv

 AI COMPARATIVE INSIGHT

This comparison explores the relationship between two distinct approaches to improving Deep Learning (DL) systems: one focused on **algorithmic logic and explainability** (Paper 1) and the other on **hardware efficiency and computational speed** (Paper 2).

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### 1. Common Themes

Despite their different focuses, the papers share several core motivations and subject matters:

**Optimization of Deep Learning:** Both papers seek to overcome the limitations of current Convolutional Neural Network (CNN) implementations. While Paper 1 optimizes for **accuracy and interpretability**, Paper 2 optimizes for **throughput and hardware efficiency**.

**Reliance on CNNs:** Both papers center on visual tasks. Paper 1 applies CNNs to satellite imagery (semantic segmentation), while Paper 2 discusses CNNs as the primary algorithm for various visual tasks.

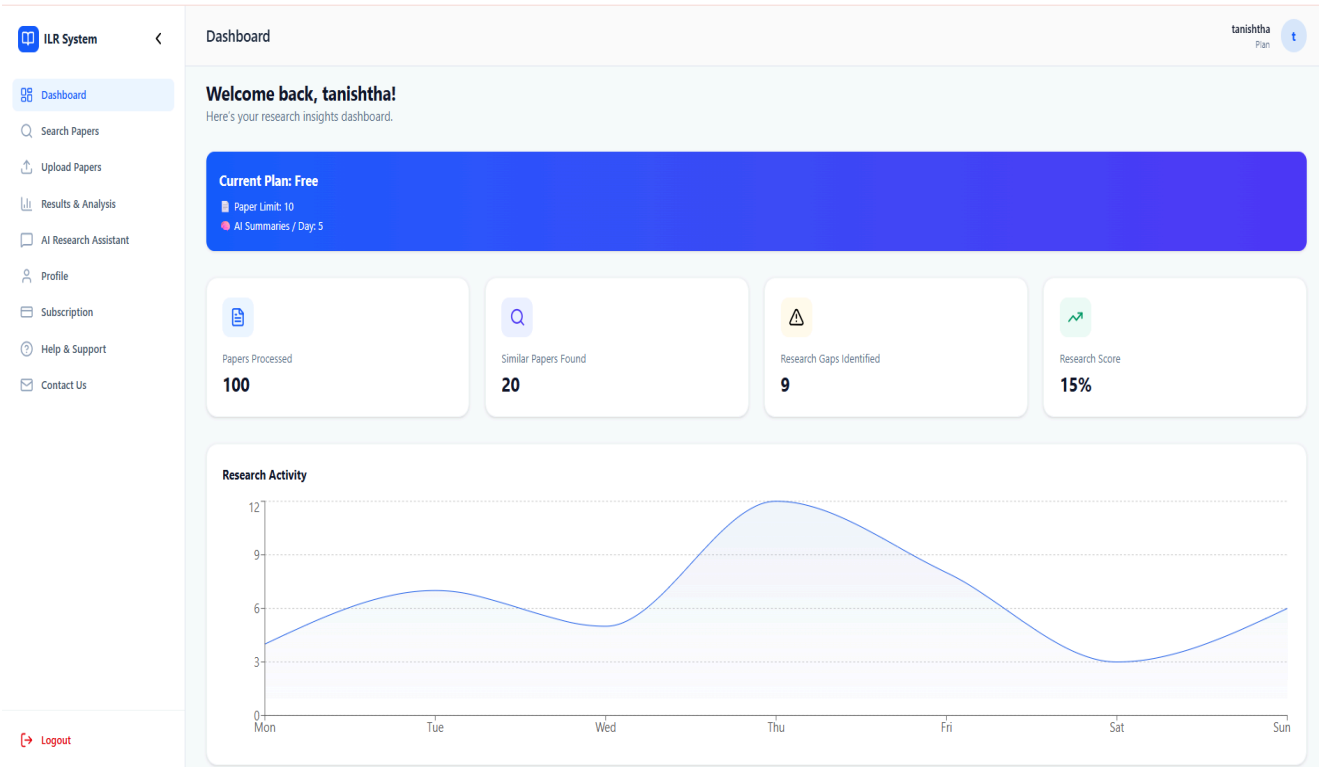
**Identification of "Systematic Shortcomings":** Both authors acknowledge that current DL frameworks are not perfect. Paper 1 identifies shortcomings in **logical output** (errors in spatial context), while Paper 2 identifies shortcomings in **resource utilization** (bottlenecks in memory bandwidth and logic resources).

**Transitioning away from "Standard" Frameworks:** Both papers suggest that the status quo (Standard Neural Networks for Paper 1; CPU/GPU-based processing for Paper 2) is insufficient for the next generation of AI applications.

# DASHBOARD OVERVIEW

Dashboard shows user activity and insights.

- Papers Processed
- Research Score
- Search History



# UPLOAD PAPERS

Users can upload PDF or ZIP files.

Steps:

1. Upload file
2. Click Summarize
3. Ask AI questions

Upload Papers

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Choose files No file chosen  
Upload PDFs or ZIP files

Daily Usage: 0 / 5

Uploaded Papers

No files uploaded

Ask AI

Ask something about uploaded papers...

Ask

# SUBSCRIPTION PLANS

The system offers multiple plans:

- Free Plan (limited access)
- Student Plan (extended access)
- Institutional Plan (unlimited access)

Subscription

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Plan

### Choose the Right Plan

Unlock advanced AI capabilities and accelerate your literature review process.

**Free Plan**  
**\$0**/month  
Perfect for individual casual researchers.

- ✓ 5 AI Summaries / day
- ✓ Basic Search
- ✓ 10 Paper Library Limit
- ✓ Standard Support

Current Plan

**Student Plan**  
**\$9**/month  
Enhanced features for dedicated students.

- ✓ Unlimited AI Summaries
- ✓ Advanced Gap Detection
- ✓ 100 Paper Library Limit
- ✓ Priority Email Support
- ✓ Export Analysis to PDF

Upgrade to Student

**Institutional Plan**  
**\$49**/month  
Full power for research labs and universities.

- ✓ Everything in Student
- ✓ Team Collaboration
- ✓ Unlimited Library
- ✓ API Access
- ✓ Dedicated Account Manager

Upgrade to Institutional

## **HELP & SUPPORT**

Users can access documentation and assistance through the Help & Support section.

